

SECTION 15816-A
FLUOROPOLYMER COATED STAINLESS STEEL DUCTWORK

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of Contract, including General and Supplementary Conditions.

1.2 DESCRIPTION OF WORK

- A. Extent of coated ductwork is indicated on drawings and in schedules, and by requirement of this specification.
- B. Refer to other specifications for ductwork accessories; not work of this specification.
- C. Refer to other specifications for fans and air handling units; not work of this specification.

1.3 QUALITY ASSURANCE:

- A. Manufacturer's Qualifications: Firms regularly engaged in the manufacture and coating of fluoropolymer coated stainless steel ductwork products of types, materials, and sizes required. Manufacturer's whose fluoropolymer coated stainless steel duct have been in satisfactory use for not less than 3 years. The owner and/or his representatives shall have the right to tour the fluoropolymer coated stainless steel ductwork manufacturer's plant anytime that fabrication is being performed on ductwork intended for this project.

1. Ductwork

- a. Base Metal shall be 304 Stainless Steel. Longitudinal seams shall be fusion welded, no filler rod used. Traverse seams shall be continuous weld. All seams shall be ground/polished smooth.
- b. Coating shall be a Teflon ETFE/ECTFE Fluoropolymer thermoplastic resin. Coating shall have an average coating thickness of 10 mils.
- c. Testing shall be performed at the request of the owner

and/or his representatives and shall be performed over the entire coated surface inside and out (where applicable) and edges. The testing shall be performed with a DC spark tester used at 250 volts/mil.

- d. Companion (Van Stone) flange is the preferred method for connecting coated duct.

- (1) Gasket: Companion flange gasket material shall be a form in place, fully expanded 100% PTFE.

- e. Heat shrink wrap shall be applied when using a slip joint. When properly joined and aligned a heat shrink band is applied around the perimeter of the joint.

2. Codes and Standards

- a. Ducts shall be listed for use without the necessity for internal fire protection sprinklers or any devices relied on the cut off air flow in the event of fire by the following:

- (1) Duct shall have a flame spread of less than 25 and a smoke generation rate of less than 50.

- b. SMACNA Standards: Comply with SMACNA's "HVAC Duct Construction Standards, Metal and Flexible" for fabrication and installation of metal ductwork and SMACNA's "Round Industrial duct construction Standards" intended for use by designers of industrial ventilation systems.

- B. Installers Qualifications: Firm with at least 3 years of successful installation experience on projects with metal ductwork, specifically industrial exhaust systems.

1. Codes and Standards:

- a. ASHRAE Standards: Comply with ASHRAE handbook, equipment Volume, Chapter 1 "Duct Construction", for fabrication and installation of metal ductwork.
- b. NFPA Compliance: Comply with NFPA 90A "Standard for the Installation of Air Conditioning and Ventilating Systems", NFPA 90B "Standard for the Installation of Warm Air Heating and Air Conditioning Systems" and NFPA 91 "Standard for the Installation of Blower and Exhaust Systems".

- c. Field Reference Manual: Have available for reference at project field office, copy of SMACNA "Round Industrial Duct Construction Standards".

1.4 SUBMITTALS:

- A. Product Data: Submit manufacturer's technical product data and installation instructions for Coated Ductwork.
- B. Shop Drawings: Submit scaled layout drawings of coated ductwork and fittings including, but not limited to, duct sizes, locations, elevations, and slopes of horizontal runs, wall and floor penetrations, and connections. Show interface and spatial relationship between coated ductwork and proximate equipment. Show modifications of indicated requirements and how those modifications ensure that free area, materials, and rigidity are not reduced.
- C. Record Drawings: At project close-out, submit record drawings of installed coated ductwork in accordance with requirements of Division 1.
- D. Maintenance Data: Submit maintenance data and parts list for coated ductwork. Include this data, shop drawings and record drawings, in maintenance manual; in accordance with requirements of Division 1.

1.5 DELIVERY, STORAGE, AND HANDLING:

- A. Protection: Protected coated ductwork from damage during shipping, storage and handling. Protection is to be applied to ends of ductwork to prevent end damage and prevent dirt and moisture from entering ducts and fittings.
- B. Delivery: Consignee must inspect shipment upon delivery and note any and all damages and discrepancies on bill of lading and notify manufacturer within 24 hours.
- C. Storage: Coated ductwork should not be stored in an area where it will have a chance to be damaged from traffic or debris. All coated ductwork should be stored on cardboard, styrofoam or similar materials. Where possible, store inside and protect from dirt and debris. Where necessary to store outside, store above grade and enclose with waterproof wrapping to protect from dirt and debris.
- D. Handling: If coating is scratched and a spark is detected, contact manufacturer for repair instructions.

PART 2 - PRODUCTS

2.1 FLUOROPOLYMER COATED STAINLESS STEEL DUCTWORK:

- A. General: Fluoropolymer coated stainless steel duct shall be constructed in full accordance with the specifications below:
1. Base metal shall be a 304 stainless steel, constructed to a duct gage and reinforcing system as per the "SMACNA Round Industrial Duct Construction Standards" with a Class 1, -6" WG schedule. Longitudinal seams shall be fusion welded, no filler rod used. Traverse seams shall be continuous weld. All seams shall be ground/polished smooth.
 2. The coating shall be a Teflon ETFE/ECTFE thermoplastic resin. The primer shall be not less than 3 mils thick. The top coat shall be 6 to 7 mils thick. The average coating thickness shall be 10 mils. Complete and uniform coating coverage is required. When coating the outside of the duct for use in chemical rooms or in environments where fumes could degrade outside of duct, the top coat of the exterior of the duct shall be 3 to 4 mils.
 3. Testing shall be performed over the entire coated surface inside and out (where applicable) and edges. The testing shall be performed with a DC spark tester used at 250 volts/mil.

2.2 MISCELLANEOUS DUCTWORK MATERIALS:

- A. General: Provide miscellaneous materials and products of types and sizes indicated and, where not otherwise indicated, provide type and size required to comply with ductwork system requirements including proper connection of ductwork and equipment.
1. Fittings:
 - a. Elbows shall have a minimum centerline radius of 1-1/2" times the duct diameter whenever possible. Unless specifically detailed otherwise, use 45 degree laterals and 45 degree elbows for branch takeoff connections. Where 90 degree branches are indicated, provide shoe tap type tees.
 2. Duct Joints:

- a. Companion Flange (Van Stone) duct joints shall be used when the ductwork is fluoropolymer coated only on the inside. The minimum number of holes for flange connections is one hole for each 6" of duct circumference to the next higher even number. In no case will flange connections have fewer than 6 holes. The minimum flange height for round ducts:

6" Dia. to 11" Dia. = 1-1/4"
12" Dia. to 24" Dia. = 1-1/2"
25" Dia. to 48" Dia. = 2"
48" Dia. And up = 2-1/2"

- (1) Nuts and Bolts: Nuts and bolts shall be minimum SAE grade 2 plated finish. Torque shall be the following:

5/16 bolts = 8 ft. lbs.
3/8 bolts = 15 ft. lbs.
7/16 bolts = 24 ft. lbs.

- (2) Companion Flange (Van Stone) joint sealant as recommended by the manufacturer. The minimum size for round ducts:

6" Dia. to 11" Dia. = 3/16"
12" Dia. to 24" Dia. = 1/4"
25" Dia. up = 3/8"

3. Ductwork Support Material: Except otherwise indicated provide matching stainless steel materials for support of ductwork.
4. Drains: Predetermine location of duct drains prior to manufacturer's fabrication. Drains are to be located in the bottom of all main and branch ducts which are P-trapped and/or as shown on the drawings to allow removal of condensate liquids. Drains are to be clear PVC pipe 1" minimum, with PVC ball valve and 2" long tail piece. Install drain lines as shown in detail in drawings.

2.3 FABRICATION:

- A. Factory-Fabricated ductwork in 4 lengths, unless otherwise indicated or required to complete runs. Pre-assemble work in shop to greatest extent possible, so as to minimize field assembly of systems. Disassemble systems only to extent necessary for shipping and handling.

- B. Factory fabricated ductwork to be of gages and reinforcement complying with SMACNA "Industrial Duct construction Standards".
- C. Factory fabricate Duct fittings to match adjoining ducts, and to comply with duct requirements as applicable to fittings. Except as otherwise indicated, fabricate elbows with center-line radius equal to associated duct width. Limit angular tapers to 30 degrees for contracting tapers and 20 degrees for expanding tapers.
- D. Factory Fabricate ductwork and fittings with predetermined location of test holes for monitors.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. General: Examine areas and conditions under which coated ductwork is to be installed. Do not proceed with work until unsatisfactory conditions have been corrected in manner acceptable to installer.

1. Installation of Coated Stainless Steel Ductwork:

- a. General: Assemble and install coated stainless steel ductwork while using extreme care not to scratch surface of coating (if scratched immediately contact manufacturer for repair instructions) and in accordance with recognized industry practices which will achieve air-tight, under 1% leakage. Install each run with minimum number of joints. Align ductwork accurately at connections, with internal surfaces smooth. Support ducts rigidly with suitable ties, braces, hangers and anchors of type which will hold duct true-to-shape and to prevent buckling. Support in accordance with SMACNA "Industrial Duct Construction Standards".
- b. Routing: Locate coating stainless steel ductwork runs, except as otherwise indicated, vertically and horizontally and avoid diagonal runs wherever possible. Locate runs as indicated by diagrams, details and notations or, if not otherwise indicated, run ductwork in shortest route which does not obstruct usable space or block access for servicing building and its equipment. Hold ducts close to walls, overhead construction, columns, and other structural and permanent enclosure elements of building. Wherever possible in finished and occupied spaces, conceal ductwork from view, by locating in mechanical shafts, hollow wall construction, or above suspended ceilings. Do not encase horizontal runs in solid partitions, except as specifically shown. Coordinate layout with suspended ceiling and lighting layouts and similar finish work.
- c. Electrical Equipment Spaces: Do not route ductwork through transformer vaults and their electrical equipment spaces and enclosures.
- d. Penetrations: Where ducts pass through interior partitions and exterior walls, and are exposed to view, conceal space between construction opening and duct with sheet metal flanges of same gauge as duct. Overlap

- opening on 4 sides by at least 1-1/2".
- e. Installation: Do not penetrate the coating for any reason. No fastening devices such as Tek screws, rivets, etc. are to be used on any part of a coated ductwork application. Test holes and slots for monitoring must be predetermined before fabrication and coating. Install coated stainless steel ductwork as shown on drawings and described herein, applicable state and city codes, and per SMACNA HVAC Duct construction Standards.

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